Everyday Context and Cognition: Links Between Household Resources and Neural Control

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Abstract

Everyday, economic and environmental conditions shape how people allocate attention and regulate behavior. We link recent household food insecurity and related socioeconomic measures to the neural indices of target-focused attention in adults using EEG gathered from a visual oddball task. We extracted P3b signals from the data of each participant, and modeled both its amplitude (the strength of a participant's attention) and latency (the evaluation speed of a participant). We tested whether food insecurity predicts these P3b measures using mixed-effect models with socioeconomic covariates, to identify contexts where associations are strongest. Across analyses, food insecurity showed diverse relationships with P3b amplitude across socioeconomic strata, while P3b latency showed a consistent pattern where greater food insecurity was associated with slower target evaluation. These results suggest resource conditions relate to core attentional and evaluative processes and motivate further work on how the socioeconomic context shape cognitive control.

1. Introduction

People allocate attention and control by weighing expected benefits against cognitive costs (Shenhav et al., 2013). Resource strain can reduce available cognitive bandwidth and alter how consistently people engage controlled processing (Mani et al., 2013; Evans & Schamberg, 2009). We test whether recent household food insecurity tracks neural indices of attention during a visual oddball task.

The target-related P3b is a centro-parietal positive ERP elicited by rare, task-relevant targets in oddball paradigms (Polich, 2007). P3b amplitude is commonly interpreted as reflecting attentional allocation and "context updating" when a goal-relevant event occurs (Donchin & Coles, 1988; Polich, 2007). More broadly, P3b amplitude is often treated as a marker of processing capacity, scaling with task engagement (Kok, 2001). Influential models also link P3b to a phasic locus coeruleus–norepinephrine (LC–NE) signal that supports updating and decision commitment (Nieuwenhuis et al., 2005). We therefore examine whether food insecurity covaries with target evaluation, quantified as target-evoked P3b amplitude.

P3b latency provides a complementary timing measure: it indexes how quickly participants complete stimulus evaluation and update task-relevant processing, often tracking evaluation/decision stages rather than response execution (Kutas et al., 1977; McCarthy & Donchin, 1981; Verleger, 1997). In classic mental-chronometry work, longer P3 latencies reflect slower stimulus-evaluation time even when overt responses are held constant (Kutas et al., 1977). If food insecurity is associated with slower or less efficient target evaluation, it should appear as delayed P3b latency even when amplitude effects vary across subgroups (Polich, 2007). We test this possibility using mixed-effects models that account for repeated trials within participants. Because amplitude differences alone cannot distinguish disengagement from altered allocation (Kok, 2001; Shenhav et al., 2013), latency adds evidence for slowed target evaluation even when amplitudes are preserved or elevated (Verleger, 1997).

This question matters scientifically because resource strain has been linked to limits in cognitive bandwidth and controlled performance, and chronic stress exposure has been tied to weaker executive resources that support stable attention (Mani et al., 2013; Evans & Schamberg, 2009; Shenhav et al., 2013). Testing whether present-day food insecurity tracks P3b amplitude and latency can clarify how everyday constraints relate to core attention and evaluation operations widely studied across cognitive neuroscience.

Finally, the rest of the paper proceeds as follows. In Section 2 we describe the dataset, variables, and our analysis plan. In Section 3 we report the main effects for the visual oddball P3b signals, and the brain–behavior linkage. In Section 4 we interpret the findings in light of prior work and outline limitations and next steps.

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Project report for the "Data Literacy" course at the University of Tübingen, Winter 2025/26 (Module ML4201). Style template based on the ICML style files 2025.

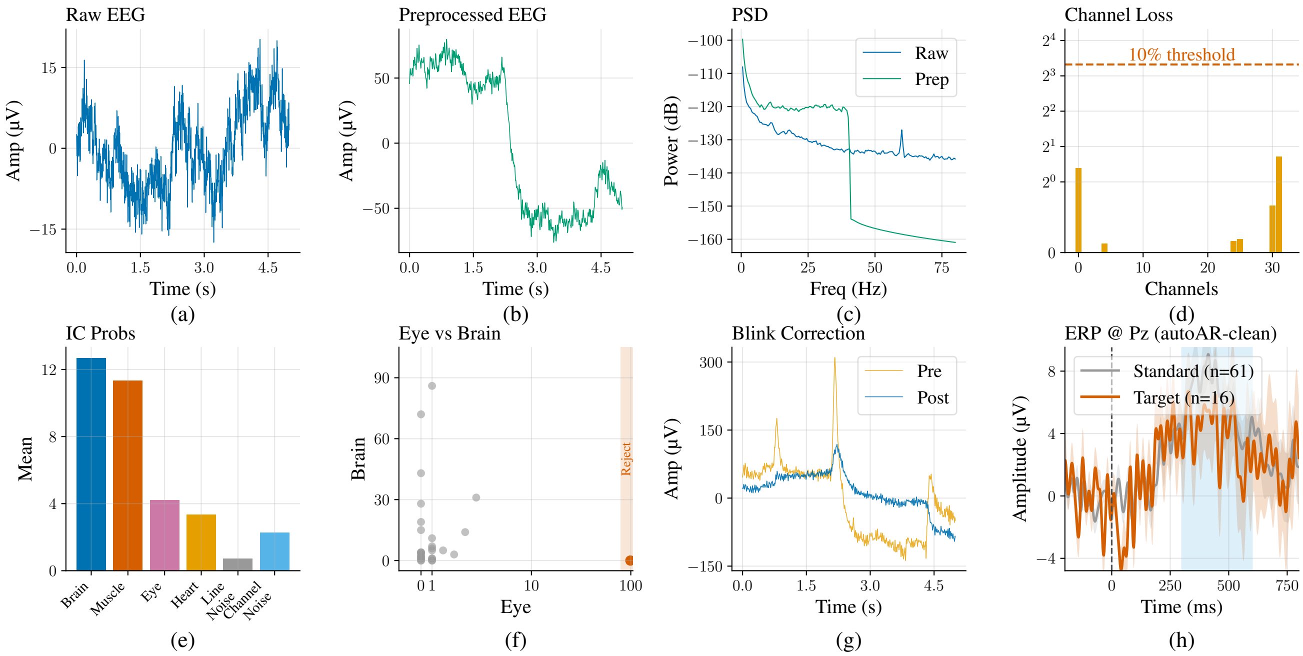

Figure 1. Preprocessing quality-control panels for the visual oddball task (shown for `sub-001`; the same workflow and checks were repeated for all participants). (a) Raw 5 s time series at Pz. (b) Preprocessed 5 s time series at Pz. (c) PSD overlay at Pz (raw vs. preprocessed). (d) Pre-ICA extreme-loss proportion by channel. (e) Mean ICLabel class probabilities across components. (f) ICLabel Eye vs. Brain scatter highlighting ocular candidates. (g) Frontal-channel blink attenuation proxy (pre vs. post ICA). (h) ERP at Pz for targets vs. standards after automated trial rejection. **Interactive Streamlit data preprocessing dashboard (designed and implemented by the authors):** `https://dataliteracy-aeardzhvggvfgwzmcvwgvh.streamlit.app/`.

2. Data and Methods

We used the public dataset "Cognitive Electrophysiology in Socioeconomic Context in Adulthood," which includes EEG from 127 adults alongside measures of food security and home/neighborhood characteristics. Participants completed ERP CORE-based tasks (active visual oddball, simple visual search, and flanker) (Kappenman et al., 2021; Isbell et al., 2025), which yield well-characterized ERPs and therefore provide a strong basis for evaluating preprocessing quality and downstream ERP interpretability.

2.1. EEG preprocessing and quality control

EEG preprocessing used an 8-stage pipeline to make continuous EEG easier to interpret while preserving time-locked responses. Recordings were standardized and filtered to reduce slow baseline drifts, suppress high-frequency noise, and narrow-band line-related structure. This is visible in Fig. 1: the baseline instability and noise in the raw segment (a) are reduced after preprocessing (b), and the power spectrum (c) becomes smoother with less high-frequency contamination and line-noise-related structure, consistent with improved signal-to-noise.

Next, we evaluated channel quality by quantifying how often each sensor observed extreme-amplitude periods ("extreme-loss"). Channels with frequent extremes can bias averages and distort scalp patterns, so they were flagged for downstream handling. In Fig. 1d, most channels show low extreme-loss (well below the illustrative 10% line), suggesting the recordings are not dominated by persistently saturated sensors, meaning our data is well-structured.

We then used independent component analysis (ICA) to remove common artifacts, especially blinks and eye movements. Components were labeled, and those with strong ocular signatures were removed. Fig. 1 also supports targeted removal: mean class probabilities in 1e are not dominated by eye labels, and the Eye-vs-Brain scatter 1f isolates a small set of high eye-likelihood components. Consistent with effective ocular correction, the frontal-channel overlay 1g shows reduced blink-like deflections after removing ocular components while leaving the remaining signal intact.

Finally, we removed the remaining artifact-contaminated trials via automated epoch rejection and computed stimulus-locked ERPs from retained epochs. The resulting Pz ERPs in 1h show a clear post-stimulus response, including the expected centro-parietal positivity in the plotted time window that indicates preserved event-locked structure after cleaning. The same preprocessing and workflow checks were applied to all participants, and could be checked in the Streamlit data preprocessing dashboard shared in Fig. 1.

2.2. Modeling approach

Our analytic sample contained 97 adults (age 21.6 ± 2.7 years; 53 female, 44 male, 1 non-binary) who completed the active visual oddball task. After automated rejection, the retained dataset comprised 7,677 trials (1,549 targets; 6,128 standards), with a median of 80 retained trials per participant (95% contributed $\geq$60). At Pz, we quantified the P3b responses for target events: (i) **latency**, defined as the timing of the P3b peak on target trials, and (ii) **amplitude**, defined as the average voltage values of the Pz EEG channel, in the 300–600 ms time window after the stimulus appears (in μV). We computed participant-level averages for description, but primary inference used trial-level models.

Food insecurity was measured with the USDA 6-item household module (FS6) and recoded into three ordered categories: 0 = high food security (FS6 = 0–1), 1 = low food security (FS6 = 2–4), and 2 = very low food security (FS6 = 5–6). In the analytic sample, 64 participants were food secure, 16 had low food security, and 17 had very low food security ($N = 97$; 1,549 target trials).

We fit trial-level mixed-effects models with a random intercept for participant to account for repeated trials. Before inference, we examined residual normality, checked heteroskedasticity via residuals versus fitted values, inspected the distribution of random intercepts, and computed the intraclass correlation to confirm non-trivial between-participant variance. These diagnostics supported the mixed-effects framework and confirmed that our variables meet the modeling assumptions for mixed-effect modeling. We also ran robustness checks to assess sensitivity to modeling assumptions. These details are provided in our GitHub repository.[1]

Primary models treated food insecurity as an ordinal predictor (0–2):

$$\text{Latency}_{ij} = \beta_0 + \beta_1 \, \text{FoodInsecurity}_j + u_j + \varepsilon_{ij}, \quad (1)$$

$$\text{Amplitude}_{ij} = \beta_0 + \beta_1 \, \text{FoodInsecurity}_j + u_j + \varepsilon_{ij}. \quad (2)$$

where i indexes trials, j indexes participants, u_j is a participant-specific random intercept, and ε_{ij} is the residual. For subgroup analyses in Fig. 2, we modeled trial-level amplitude using continuous FS6 (0–6), adjusted for subgroup differences with covariates (subjective SES and, where applicable, education), and included a target-status interaction when estimating target–standard differences.

3. Results

We first report patterns of association between food insecurity (FS6) and P3b amplitude (Fig. 2), followed by the latency analysis testing of dose–response effects on stimulus evaluation speed (Fig. 3).

[1] https://github.com/mhdihso/DATA_LITERACY

3.1. Food Security on P3b Amplitude

Figure 2 summarizes how food insecurity (FS6; 0–6 scale) relates to P3b amplitude, interpreted as the attention allocated to task events. To address repeated measurements (many trials per person), we fit trial-level linear mixed-effects models with a random intercept for participant. Rather than a single pooled effect, the figure highlights heterogeneity across selected subgroups. We observe that in smaller households, higher food insecurity relates to a stronger attention signal (P3b) by participants ($b = +1.294$ μV per FS6 point, 95% CI $[+0.544, +2.043]$, $p = 0.000714$; 356 trials). A similar positive association appeared in the bachelor's-degree subgroup ($b = +1.129$ μV, 95% CI $[+0.080, +2.179]$, $p = 0.0350$; 307 trials).

In contrast, participants with no college degree showed a negative (but less significant) association between food insecurity and attention relevant P3b signals ($b = -0.663$ μV, 95% CI $[-1.395, +0.070]$, $p = 0.0762$; 330 trials), suggesting a possible reduction in target-related attention with increasing food insecurity in that subgroup. Finally, to index attentional selectivity, we modeled the target–standard stimuli difference for high household-incomes and observed that in the highest income families, higher food insecurity predicted a larger target–standard separation ($b = +1.204$ μV per FS6 point, 95% CI $[+0.198, +2.211]$, $p = 0.0190$; 2131 trials), indicating a stronger difference in attention of target and non-target trials as food insecurity increases.

3.2. Food Security on P3b Latency

To test whether food insecurity relates to stimulus evaluation speed, we fit a trial-level linear mixed-effects model predicting P3b latency from food insecurity category. Each one-category increase in food insecurity was associated with an estimated 9.3 ms increase in P3b latency ($\beta = 9.33$ ms, 95% CI [0.56, 18.09], $p = 0.037$). This aligns with the dose–response pattern in Figure 3 and shows that the average the evaluation speed of participant (latency) increases as food insecurity decreases (417.6 ms food secure; 423.7 ms low food security; 436.8 ms very low food security).

Table 1. Covariate robustness of the association between food insecurity and P3b latency (complete cases across all covariates: 94 subjects, 1,503 trials).

Model	β (ms)	p-value	95% CI
M0: Unadjusted	9.75	0.032	[0.8, 18.7]
M1: + Age	10.11	0.028	[1.1, 19.1]
M2: M1 + Gender	9.60	0.031	[0.9, 18.3]
M3: M2 + Education	9.29	0.037	[0.5, 18]
M4: M3 + SubjSES	10.16	0.038	[0.6, 19.8]
M5: M4 + Parent Ed	10.59	0.031	[1.0, 20.2]

* $p < .05$

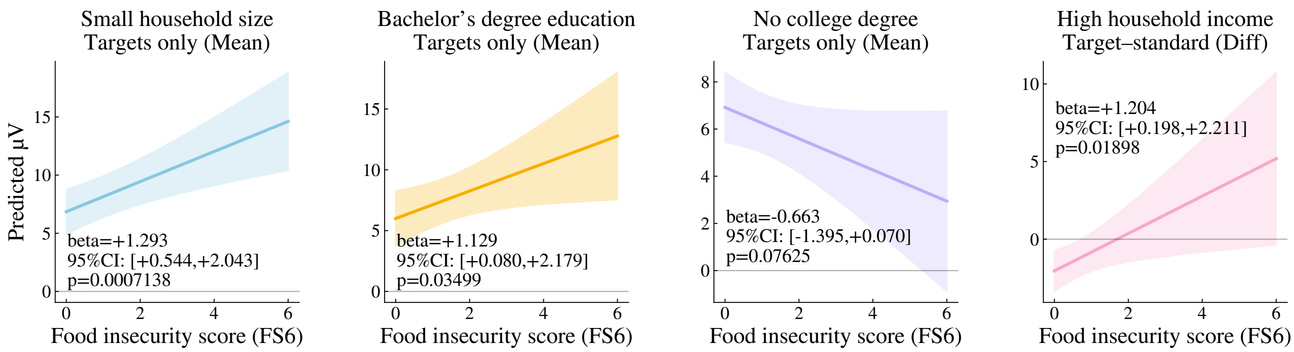

Figure 2. Food insecurity (FS6) versus predicted P3b amplitude (attentional allocation), estimated from trial-level linear mixed-effects models with a random intercept for participant. Shaded bands show 95% confidence intervals for the fixed-effects predictions.

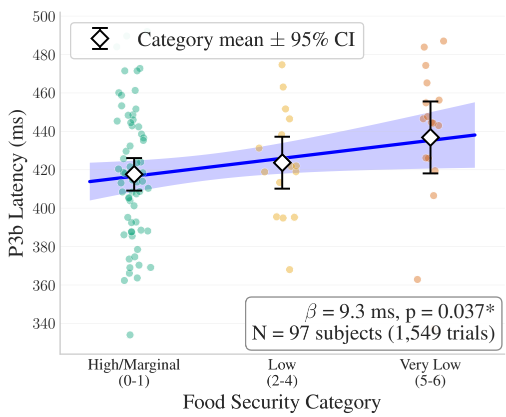

Figure 3. Relationship between food security and P3b latency.

To evaluate robustness to potential confounding, we refit mixed-effects models with covariates added sequentially (age, gender, education, subjective SES, parent education). The food insecurity coefficient remained stable across specifications (roughly 9–11 ms per category), with similar p-values and overlapping confidence intervals (Table 1), suggesting the association was not explained by these measured covariates. As an additional check, a participant-level OLS model with HC3 robust standard errors on mean latencies yielded a comparable slope ($\beta = 9.14$ ms, 95% CI [0.2, 18.1], $p = 0.0448$), indicating the conclusion is not driven by unequal trial counts across participants.

4. Discussion & Conclusion

Food insecurity is strongly correlated with socioeconomic resources and captures everyday material strain. Consistent with our motivation to test whether material strain relates to both attentional allocation (P3b amplitude) and stimulus evaluation speed (P3b latency), we analyzed EEG from a visual oddball task in 97 adults. Higher food insecurity was associated with delayed P3b latency to targets, with each category increase linked to about 9 ms slower evaluation (d=0.37). This association was robust to adjustment for age, gender, education, and subjective SES, and it replicated in a participant-level analysis of mean latencies, suggesting it is not driven by unequal trial counts across participants.

A plausible interpretation is consistent with cognitive bandwidth accounts of scarcity-related strain (Mani et al., 2013): ongoing resource management may increase cognitive load and reduce processing efficiency, showing as slower stimulus evaluation. In contrast, amplitude results were not uniform. Exploratory subgroup models suggested that food insecurity related to target-evoked amplitude and/or target–standard differentiation in ways that varied across socioeconomic contexts, pointing to context-dependent shifts in attentional allocation rather than a single global effect.

Because these amplitude patterns were examined across multiple subgroup specifications, they should be treated as hypothesis-generating rather than confirmatory and may be sensitive to multiple testing. Future work should preregister a smaller set of interaction tests, connect neural measures to behavior (accuracy and reaction time), and test generalization across tasks and samples, ideally longitudinally.

Limitations. Our trial-level observations were clustered within participants, so basic trial-wise regression methods fail in their assumptions and can understate uncertainty. To deal with this, we used mixed-effects models to account for within-subject dependence while retaining trial-level information; however, subgroup estimates remain less precise. Additionally, this study is cross-sectional and observational, so we can't draw causal conclusions, and unmeasured factors (e.g., sleep and stress) may still contribute to the observed associations.

Overall, our results complement prior work linking socioeconomic conditions to P3b amplitude (Isbell et al., 2025) by highlighting a new, robust association between the food insecurity and neural signals. Taken together, the findings suggest that material strain may be reflected primarily in slower stimulus evaluation, alongside heterogeneous attentional allocation patterns.

Contribution Statement

Abtin Mogharabin went through the past literature for the initial literature review, worked on 6 scripts of the visual oddball preprocessing, and assisted in the mixed effect modeling of P3b amplitudes. Kourosh Sharifi went through the data files, detected and fixed the issue with the header files, conducted a detailed EDA, worked on 2 scripts of visual oddball preprocessing, and helped with the website creation. Seyedmehdi Hosseini assisted in EDA, worked on 2 scripts of visual oddball preprocessing, and developed the website. Mina Mikhael assisted to check on the structure of our EEG files and preprocessing, worked on the visual oddball latency analysis with mixed-effect models, assisted with the visualizations and worked on 2 scripts of visual oddball preprocessing. After these, all members collaborated together to write the full report.

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